

Composite divisor parametrizations for the Erdős–Straus equation

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Abstract

We give exact divisor forms of the two standard parametrizations of Type I and Type II solutions of the Erdős–Straus equation for a prime denominator. Composite divisors are retained throughout. We prove a two-sided residue criterion for a divisor of $p + k$, classify all Type II parameters using a common prime factor outside an explicit finite range, and derive a congruence criterion for combining factors arising from different shifts. These identities produce additional families of solutions, but do not prove the Erdős–Straus conjecture.

1 Introduction

The Erdős–Straus conjecture asks whether every integer $n \geq 2$ admits

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad x, y, z \in \mathbb{N}.$$

It is enough to consider prime p . For prime p , every solution is of Type I or Type II according as p divides one or two denominators. The purpose of this paper is algebraic: we record several exact consequences of the divisor parametrizations and isolate the part of the search that is genuinely arithmetic.

2 The two divisor criteria

Proposition 2.1 (Pomerance–Weingartner, specialized to (m=4)). *Let p be prime. If there are $u, d, f \in \mathbb{N}$ such that*

$$f \mid 4u^2d + 1, \quad 4ud \mid p + f,$$

then $4/p$ has a Type I decomposition. Conversely, every Type I decomposition gives such a triple.

Proof. This is Corollary 2.2 of Pomerance–Weingartner [1] with numerator $m = 4$ and prime denominator $n = p$. That result states precisely that the two displayed divisibilities are necessary and sufficient for a Type I solution; no additional coprimality condition is needed when p is prime. $\square$

Proposition 2.2 (Pomerance–Weingartner, specialized to (m=4)). *Let p be prime. If there are $a, b, e \in \mathbb{N}$ such that*

$$e \mid a + b, \quad 4ab \mid p + e,$$

then $4/p$ has a Type II decomposition. Conversely, every Type II decomposition gives such a triple.

Proof. This is Corollary 2.4 of Pomerance–Weingartner [1] with numerator $m = 4$ and prime denominator $n = p$. $\square$

3 A Type I divisor construction

Theorem 3.1. *Let $p \equiv 1 \pmod{4}$ be prime and let $k \geq 1$. Suppose*

$$t \mid p+k, \quad t \equiv -1 \pmod{k}, \quad e = \frac{t+1}{k}, \quad f = \frac{p+k}{t},$$

and $ef \equiv 1 \pmod{4}$. Then $4/p$ has a Type I decomposition.

Proof. Set $d = (ef - 1)/4$. Then $f \mid 4d + 1$ and

$$p + f = ft - k = f(t + 1) - k = fke - k = k(ef - 1) = 4kd.$$

Thus $4d \mid p + f$ and Proposition 2.1 applies with $u = 1$. □

Corollary 3.2. *If k is even and $q \mid p + k$ is prime, then*

$$q \equiv 3k - 1 \pmod{4k}$$

implies a Type I decomposition.

Proof. Take $t = q$. Then $(q + 1)/k \equiv 3 \pmod{4}$. If $4 \mid k$, then $p + k \equiv 1 \pmod{4}$ and $q \equiv 3 \pmod{4}$; if $k \equiv 2 \pmod{4}$, then $p + k \equiv 3 \pmod{4}$ and $q \equiv 1 \pmod{4}$. In both cases $f = (p + k)/q \equiv 3 \pmod{4}$, so Theorem 3.1 applies. □

4 Type II parameters using a common prime factor

Write $q = 4m - 1$ and suppose $q \mid p + k$. A Type II divisor using this same q must be a divisor $h \mid m^2$ with $q \mid p + 4h$.

Theorem 4.1. *Assume $q \geq k$. Every such h is one of the following:*

$$h = km, \quad h = \frac{k}{4} \ (4 \mid k), \quad \text{or} \quad q \leq \frac{4r(r-1)}{k-4r} - 1 \quad \text{for some } 1 \leq r < k/4.$$

Proof. Since $p \equiv -k \pmod{q}$ and $4^{-1} \equiv m \pmod{q}$, the congruence $q \mid p + 4h$ is equivalent to $h \equiv km \pmod{q}$. Write

$$h = km + \ell q = (k + 4\ell)m - \ell.$$

If $\ell \geq 1$, then $h > 4\ell m$. Let $g = (h, m) = (\ell, m)$. From $h \mid m^2$, writing $h = gh_1$, $m = gm_1$ with $(h_1, m_1) = 1$ gives $h_1 \mid g$, hence $h \leq g^2 \leq \ell^2$, a contradiction.

Let $\ell = -r < 0$. Positivity gives

$$r(4m - 1) < km.$$

Since $q = 4m - 1 \geq k$, we have $r < m$. If $4r > k$, then $(4r - k)m < r$, which is impossible because $r < m$ and $4r - k \geq 1$. Hence $4r \leq k$. Equality gives $k = 4r$ and $h = r = k/4$; otherwise $k - 4r > 0$, and $h = (k - 4r)m + r$. The same argument gives $h \leq r^2$, so with $a = k - 4r$,

$$am + r \leq r^2, \quad m \leq \frac{r(r-1)}{a}.$$

The case $\ell = 0$ gives $h = km$. □

Corollary 4.2. *If $q \equiv 3 \pmod{8}$ divides $p + 2$, then no divisor $h \mid ((q + 1)/4)^2$ satisfies $q \mid p + 4h$.*

Proof. Here $k = 2$ and $m = (q + 1)/4$ is odd. The alternatives $h = km$ and $h = k/4$ are impossible, and $q < 1$ is impossible. □

5 Combining different shifts

Lemma 5.1. *Let $e \mid t + 1$, $k = (t + 1)/e$, and $t \mid p + k$. If $q \mid t$, then for every integer s ,*

$$q \mid p + s \iff q \mid es - 1.$$

Proof. Modulo q , $p \equiv -k$ and $ek = t + 1 \equiv 1$. Therefore $q \mid p + s$ is equivalent to $k \equiv s$, which is equivalent to $es \equiv 1$. $\square$

Proposition 5.2. *Let $p \equiv 1 \pmod{24}$ be prime and let $a \equiv 2 \pmod{3}$. If a prime $q \neq 3$ divides $ap + 1$ and either*

$$q \equiv -1 \pmod{4a} \quad \text{or} \quad 3q \equiv -1 \pmod{4a},$$

then $4/p$ has a Type II decomposition.

Proof. Since $3 \mid ap + 1$, set $s = q$ in the first case and $s = 3q$ in the second. Then $s \mid ap + 1$ and $s \equiv -1 \pmod{4a}$. Proposition 2.2, with $u = 1$, gives the result. $\square$

Theorem 5.3. *Let $p \equiv 1 \pmod{24}$ be prime. Consider the Type II criterion with $u = 1$.*

1. *For $a = 2$, a witness exists if and only if the odd integer*

$$N_2 = \frac{2p + 1}{3}$$

has a prime divisor congruent to 5 or 7 (mod 8).

2. *For $a = 3$, a witness exists if and only if, among the prime divisors of*

$$N_3 = \frac{3p + 1}{4},$$

there is either a divisor congruent to 11 (mod 12), or there are divisors congruent to 5 and 7 (mod 12), respectively.

Consequently, failure of both $a = 2$ and $a = 3$ is equivalent to the two explicit conditions

$$q \mid N_2 \implies q \equiv 1, 3 \pmod{8},$$

and

no $q \mid N_3$ is 11 (mod 12), and not both classes 5, 7 (mod 12) occur.

Proof. For $a = 2$, the criterion asks for a divisor $s \mid 2p + 1$ with $s \equiv 7 \pmod{8}$. Since $2p + 1 = 3N_2$, $N_2 \equiv 1 \pmod{8}$, and the group $(\mathbb{Z}/8\mathbb{Z})^\times$ has exponent two, the signed divisor set is the subgroup generated by the residue classes of the prime divisors of $2p + 1$. The factor 3 is always present. This subgroup contains 7 exactly when another prime divisor has class 5 or 7, because $3 \cdot 5 \equiv 7 \pmod{8}$. This proves (1).

For $a = 3$, the criterion asks for an odd divisor $s \mid 3p + 1$ with $s \equiv 11 \pmod{12}$. Since $3p + 1 = 4N_3$, the odd divisors are those of N_3 , and N_3 is coprime to 6. Every element of $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$ has order two. Hence the signed divisor set is the subgroup generated by the prime residue classes. It contains 11 exactly if one generator is 11, or if generators in both classes 5 and 7 occur, since $5 \cdot 7 \equiv 11 \pmod{12}$. This proves (2) and the final equivalence. $\square$

6 Conclusion

The results above enlarge the explicit Type I and Type II families and give finite classifications for several same-factor constructions. They do not imply that every prime is covered. In particular, the remaining assertion that every prime has a Type II representation is a separate open problem.

References

- [1] C. Pomerance and A. Weingartner, *Exceptions to the Erdős–Straus–Schinzel conjecture*, Ramanujan J. 69 (2026), Article 31.
- [2] M. Chamberland, *The Erdős–Straus conjecture and the structure of primes*, INTEGERS 26 (2026), A42.